



## Sydney Girls High School

### 2021 Alternate Task 4

# Mathematics Extension 1

---

#### General

#### Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using a black pen
- Calculators approved by NESA may be used
- A NESA reference sheet has been provided for use
- For questions in Section II, show relevant mathematical reasoning and/or calculations

---

#### Total marks :

**40**

#### Section 1 – 10 marks (pages 2 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

#### Section II – 30 marks (pages 7 – 12)

- Attempt Questions **11 – 13**
- Allow about 45 minutes for this section

#### **THIS IS NOT A TRIAL PAPER**

It does not reflect the format or the content of the 2021 HSC Examination paper  
in this subject.

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

For each question, select the correct response A, B, C or D.

List the correct response only on YOUR writing paper for questions 1 – 10.

---

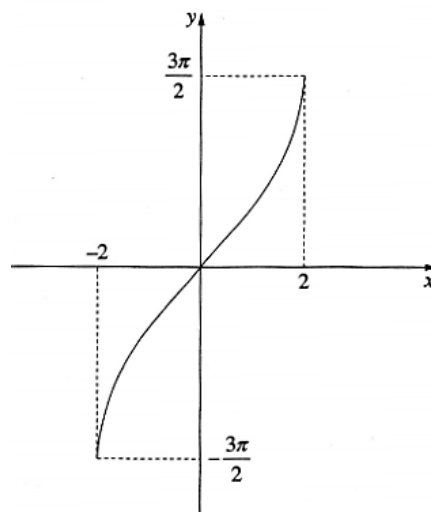
1. Which function best describes the following graph?

(A)  $y = 3 \sin^{-1} 2x$

(B)  $y = \frac{3}{2} \sin^{-1} 2x$

(C)  $y = 3 \sin^{-1} \frac{x}{2}$

(D)  $y = \frac{3}{2} \sin^{-1} \frac{x}{2}$



2. In the cartesian plane, a vector perpendicular to the line  $3x + 2y + 1 = 0$  is

(A)  $3\vec{i} + 2\vec{j}$

(B)  $-\frac{1}{2}\vec{i} + \frac{1}{3}\vec{j}$

(C)  $2\vec{i} - 3\vec{j}$

(D)  $\frac{1}{2}\vec{i} - \frac{1}{3}\vec{j}$

3. Which group of three numbers could be the roots of the polynomial equation

$$x^3 + px^2 - 26x + 24 = 0 ?$$

(A) 2, 3, 4

(B) 1, -6, 4

(C) -1, -2, 12

(D) -1, -3, -8

4. 
$$\frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}} =$$

(A)  $\cos \theta$

(B)  $\sec \theta$

(C)  $\tan \theta$

(D)  $\cot \theta$

5. Which expression is equal to  $\int \cos^2 \frac{2x}{5} dx$ ?

(A)  $\frac{x}{2} - \frac{2}{5} \sin \frac{4x}{5} + C$

(B)  $\frac{x}{2} + \frac{2}{5} \sin \frac{4x}{5} + C$

(C)  $\frac{x}{2} - \frac{5}{8} \sin \frac{4x}{5} + C$

(D)  $\frac{x}{2} + \frac{5}{8} \sin \frac{4x}{5} + C$

6. In the expression  $(2x + k)^6$  the coefficients of  $x$  and  $x^2$  are equal.

What is the value of  $k$ ?

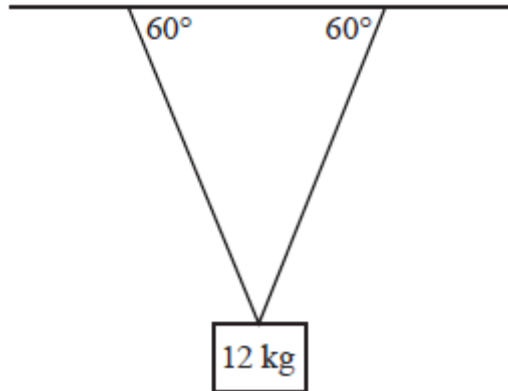
- (A) 5
- (B) 6
- (C) 11
- (D) 12

7. A group consisting of two adults, two boys and two girls is to be seated at a round table. The adults are to be seated together. The boys and girls are to sit in alternating positions? How many different seating arrangements are possible?

- (A) 8
- (B) 16
- (C) 24
- (D) 30

8. A 12kg mass is suspended in equilibrium from a horizontal ceiling by two identical light strings. Each string makes an angle of  $60^\circ$  with the ceiling, as shown. If the force of gravity is  $g \text{ m/s}^2$  then the magnitude, in newtons, of the tension in each string is equal to:

- (A)  $12g$
- (B)  $24g$
- (C)  $4\sqrt{3}g$
- (D)  $8\sqrt{3}g$

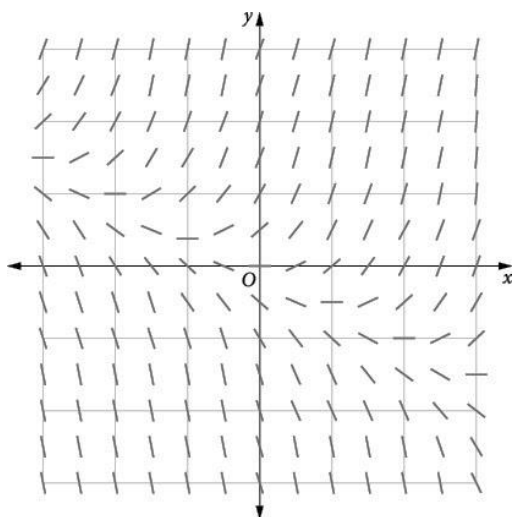


9. The volume  $V$  of a spherical balloon of radius  $r$  mm is increasing at a constant rate of  $800 \text{ mm}^3$  per second. What is the rate of change of the radius with respect to time?

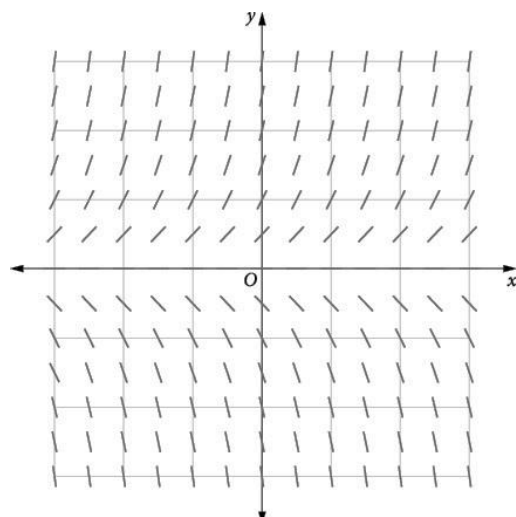
- (A)  $\frac{\pi r}{100} \text{ mm/second}$
- (B)  $\frac{100}{\pi r} \text{ mm/second}$
- (C)  $\frac{\pi r^2}{200} \text{ mm/second}$
- (D)  $\frac{200}{\pi r^2} \text{ mm/second}$

10. Which of the following direction fields could have the differential equation  $\frac{dy}{dx} = x - ky$  as a solution if  $k > 0$  ?

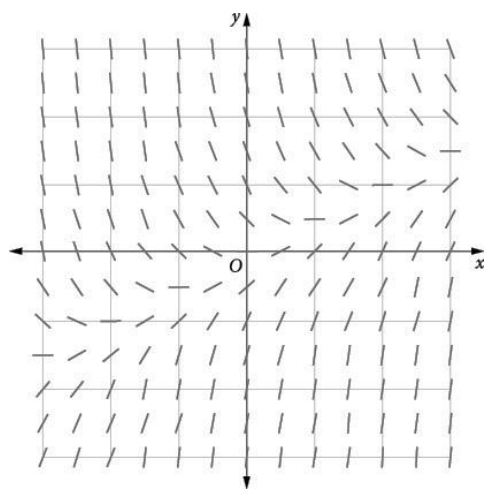
(A)



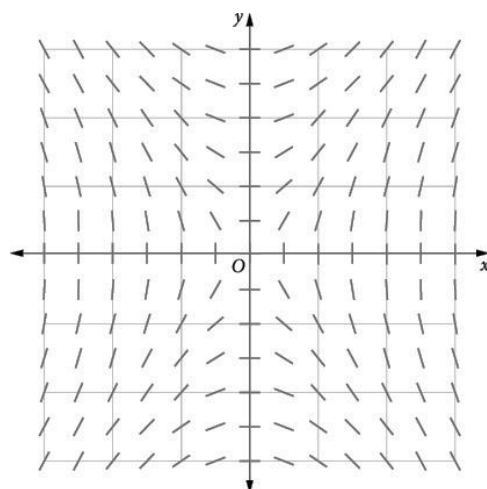
(B)



(C)



(D)



## Section II

30 marks

Attempt Questions 11 – 13

Allow about 45 minutes for this section

Begin a new page for each question.

Your responses should include relevant mathematical reasoning and/or calculations.

---

### Question 11 Begin a new Page

(10 marks)

(a)

(i) Show that  $(x - 3)$  is a factor of the polynomial

1

$$P(x) = x^3 - 4x^2 + x + 6.$$

(ii) Hence, express  $P(x)$  in factored form.

2

(b) Evaluate  $\lim_{x \rightarrow 0} \left( \frac{\sin \frac{x}{3}}{4x} \right)$

1

(c) Solve:  $\frac{x^2 - 9}{3x} > 0$

2

(d) Evaluate:

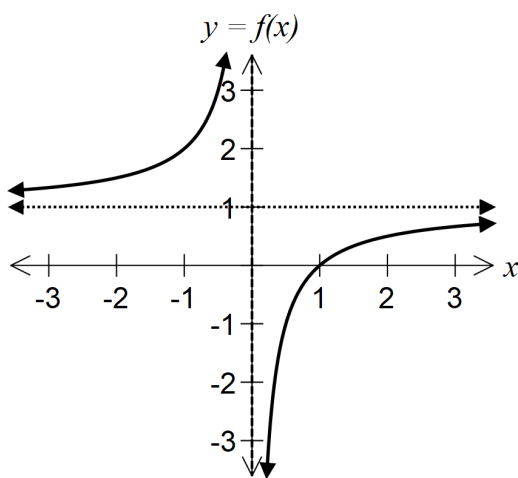
2

$$\int \frac{2x}{\sqrt{25 - x^2}} dx$$

Question 11 continues on the next page.

**Question 11** (continued)

(e) Consider the function  $f(x) = \frac{x-1}{x}$ , which is graphed below:



Using the graph of  $y = f(x)$ , sketch the function  $y = \{f(x)\}^2$ .

**2**

**End of Question 11**

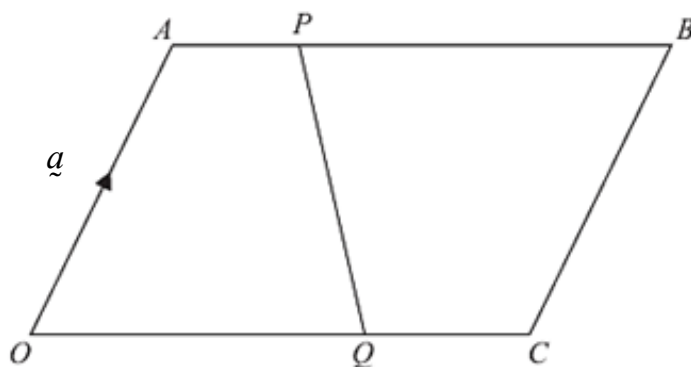
**Question 12** Begin a new page.**(10 marks)**

- (a) In the diagram below,  $OABC$  is a parallelogram.  $\overrightarrow{OA} = \underline{a}$  and  $\overrightarrow{OC} = \underline{c}$  **3**

$P$  is the point on  $AB$  such that  $AP = \frac{1}{4}AB$ .

$Q$  is the point on  $OC$  such that  $OQ = \frac{2}{3}OC$ .

Find  $\overrightarrow{PQ}$  in terms of  $\underline{a}$  and  $\underline{c}$ , giving your answer in simplest form.



- (b) Find the volume, in terms of  $\pi$ , of the solid formed when the area **3**  
bounded by the curve  $y = \frac{1}{2}\log_e x$ , the lines  $y = -1$ ,  $y = 2$  and the  $y$ -axis  
is rotated about the  $y$ -axis.

**Question 12 continues on the next page.**

**Question 12** (continued)

(c) Consider the expansion of  $(1+x)^{n-1}$  for integers  $n > 2$  .

(i) Show that:

$$n\binom{n-1}{1} + n\binom{n-1}{2} + \dots + n\binom{n-1}{n-2} = n(2^{n-1} - 2) \quad \mathbf{3}$$

(ii) Find the smallest positive integer  $n$  such that:

$$n\binom{n-1}{1} + n\binom{n-1}{2} + \dots + n\binom{n-1}{n-2} > 5000 \quad \mathbf{1}$$

**End of Question 12**

**Question 13** Begin a new page.**(10 marks)**

- (a) A particle is moving in a straight line and is oscillating backwards and forwards. At time  $t$  seconds, it has displacement  $x$  metres from a fixed point  $O$  on the line, where  $x = A \cos\left(\frac{\pi}{4}t + \alpha\right)$ ,  $A > 0$ , and  $0 < \alpha < \frac{\pi}{2}$ . After 1 second the particle is 2 metres to the right of  $O$ , and after 3 seconds it is 4 metres to the left of  $O$ .

Given that  $A \cos \alpha - A \sin \alpha = 2\sqrt{2}$  and  $A \cos \alpha + A \sin \alpha = 4\sqrt{2}$ :

- (i) Solve the equations simultaneously, to find the values of  $A$  and  $\alpha$  in exact form. **2**
- (ii) Show that the particle first passes through  $O$  after  $\frac{4}{\pi} \tan^{-1} 3$  seconds. **2**

**Question 13 continues on the next page.**

### Question 13 (continued)

- (b) The position coordinates of any point on the path of a projectile at time  $t \geq 0$  in seconds, with initial velocity  $v \text{ ms}^{-1}$  at an angle of projection  $\theta$ , and acceleration downwards due to gravity,  $g$ , are:

$$x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - \frac{1}{2}gt^2$$

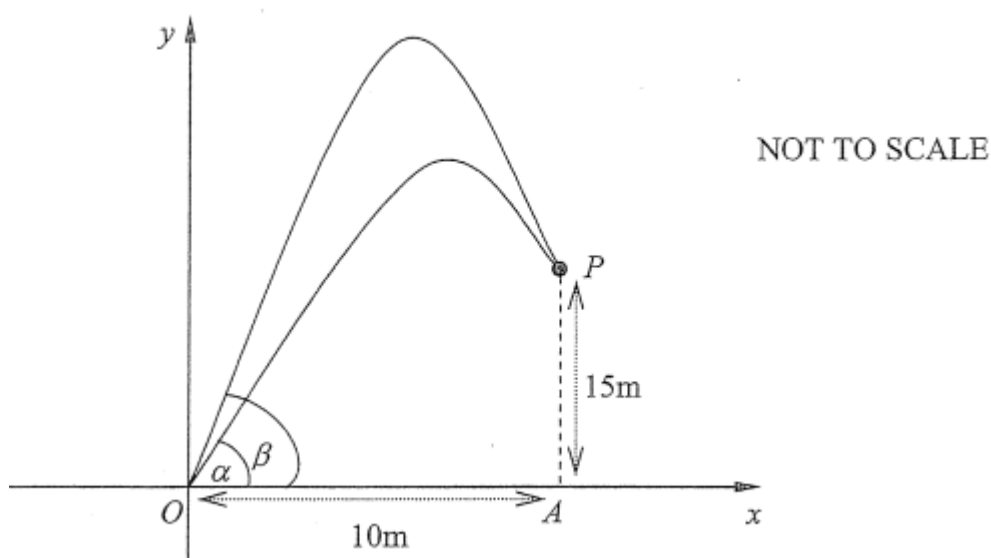
- (i) Show that the equation of the path of a projectile is given by:

2

$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

Harry throws a tennis ball from a fixed point  $O$  on level ground, with a velocity  $v = 7\sqrt{10} \text{ ms}^{-1}$  at an angle  $\beta$  with the horizontal. Shortly afterwards he throws another tennis ball from the same point at the same speed but at a different angle to the horizontal,  $\alpha$ , where  $\alpha < \beta$  as shown.

The two tennis balls collide at a point  $P$ , vertically above the point  $A$  on the ground, where  $OA = 10 \text{ m}$  and  $AP = 15 \text{ m}$ . The acceleration downwards due to gravity is  $g = 9.8 \text{ ms}^{-2}$ .



- (ii) Show that  $\tan \alpha = 2$  and  $\tan \beta = 8$ .
- (iii) Find, in surd form, the time elapsed between when the tennis balls were thrown.

2

2

**End of paper**



SOLUTIONS

**Sydney Girls High School**

**2021 Alternate Task 4**

# Mathematics Extension 1

---

**General  
Instructions**

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using a black pen
- Calculators approved by NESA may be used
- A NESA reference sheet has been provided for use
- For questions in Section II, show relevant mathematical reasoning and/or calculations

---

**Total marks :  
40**

**Section 1 – 10 marks (pages 3– 6)**

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

**Section II – 30 marks (pages 7 – 12)**

- Attempt Questions 11 – 13
- Allow about 45 minutes for this section

**THIS IS NOT A TRIAL PAPER**

It does not reflect the format or the content of the 2021 HSC Examination paper  
in this subject.

## 2021 Extension 1 Task 4

### Multiple choice

1. C

6. A

2. A

7. B

3. B

8. C

4. A

9. D

5. D

10. C

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

For each question, select the correct response A, B, C or D.

List the correct response only on YOUR writing paper for questions 1 – 10.

$$D: -1 \leq \left(\frac{x}{2}\right) \leq 1$$

$$-2 \leq x \leq 2$$

$$R: -\frac{\pi}{2} \leq \left(\frac{y}{3}\right) \leq \frac{\pi}{2}$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

$$\therefore \frac{y}{3} = \sin^{-1} \frac{x}{2}$$

$$y = 3 \sin^{-1} \frac{x}{2}$$

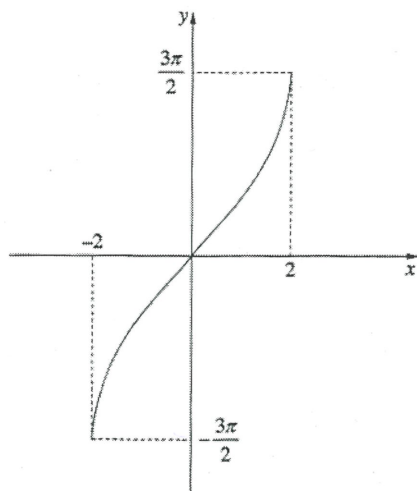
1. Which function best describes the following graph?

(A)  $y = 3 \sin^{-1} 2x$

(B)  $y = \frac{3}{2} \sin^{-1} 2x$

(C)  $y = 3 \sin^{-1} \frac{x}{2}$

(D)  $y = \frac{3}{2} \sin^{-1} \frac{x}{2}$



2. In the cartesian plane, a vector perpendicular to the line  $3x + 2y + 1 = 0$  is

(A)  $3\vec{i} + 2\vec{j}$

(B)  $-\frac{1}{2}\vec{i} + \frac{1}{3}\vec{j}$

(C)  $2\vec{i} - 3\vec{j}$

(D)  $\frac{1}{2}\vec{i} - \frac{1}{3}\vec{j}$

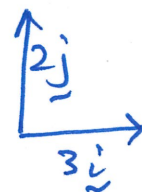
$$3x + 2y + 1 = 0:$$

$$2y = -3x - 1$$

$$y = -\frac{3}{2}x - \frac{1}{2}$$

$$m_L = -\frac{3}{2}$$

$$\therefore m_V = \frac{2}{3}$$



$$\therefore 3\vec{i} + 2\vec{j}$$

3. Which group of three numbers could be the roots of the polynomial equation

$$x^3 + px^2 - 26x + 24 = 0?$$

(A) 2, 3, 4

(B) 1, -6, 4

(C) -1, -2, 12

(D) -1, -3, -8

$$x^3 + \overset{-b}{p}x^2 + \overset{+c}{-26}x + \overset{-d}{24} = 0.$$

$$\alpha\beta\gamma = -24$$

$$2 \times 3 \times 4 \neq -24 \therefore \text{not (A)}$$

$$-1 \times -2 \times 12 \neq -24 \therefore \text{not (C)}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -26$$

$$1 \times -6 + 1 \times 4 + -6 \times 4 = -6 + 4 - 24 = -26 \therefore \text{(B)}$$

4.

$$\frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}} =$$

(A)  $\cos \theta$

(B)  $\sec \theta$

(C)  $\tan \theta$

(D)  $\cot \theta$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\Rightarrow \frac{\frac{1}{t} - t}{\frac{1}{t} + t}$$

$$= \frac{1 - t^2}{t} \div \frac{1 + t^2}{t}$$

$$= \frac{1 - t^2}{t} \times \frac{t}{1 + t^2}$$

$$= \frac{1 - t^2}{1 + t^2} = \cos \theta \quad \text{(A)}$$

5. Which expression is equal to  $\int \cos^2 \frac{2x}{5} dx$ ?

(A)  $\frac{x}{2} - \frac{2}{5} \sin \frac{4x}{5} + C$

(B)  $\frac{x}{2} + \frac{2}{5} \sin \frac{4x}{5} + C$

(C)  $\frac{x}{2} - \frac{5}{8} \sin \frac{4x}{5} + C$

(D)  $\frac{x}{2} + \frac{5}{8} \sin \frac{4x}{5} + C$

$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x.$$

$$\cos^2 \left( \frac{2x}{5} \right) = \frac{1}{2} + \frac{1}{2} \cos \left( 2 \times \frac{2x}{5} \right)$$

$$\cos^2 \left( \frac{2x}{5} \right) = \frac{1}{2} + \frac{1}{2} \cos \left( \frac{4x}{5} \right)$$

$$\therefore \int \left( \frac{1}{2} + \frac{1}{2} \cos \left( \frac{4x}{5} \right) \right) dx$$

$$= \frac{1}{2}x + \frac{1}{2} \times \frac{5}{4} \sin \frac{4x}{5}$$

$$= \frac{1}{2}x + \frac{5}{8} \sin \frac{4x}{5} + C$$

6. In the expression  $(2x + k)^6$  the coefficients of  $x$  and  $x^2$  are equal.

What is the value of  $k$ ?

(A) 5

(B) 6

(C) 11

(D) 12

$$(2x + k)^6 = {}^6C_0 (2x)^6 + {}^6C_1 (2x)^5 k + {}^6C_2 (2x)^4 k^2 \\ + {}^6C_3 (2x)^3 k^3 + {}^6C_4 (2x)^2 k^4 + {}^6C_5 (2x) k^5 + {}^6C_6 (2x)^0 k^6.$$

$$\text{coeff of } x = \text{coeff of } x^2$$

$${}^6C_5 \times 2 \times k^5 = {}^6C_4 \times 2^2 \times k^4$$

$${}^6C_5 \times k = {}^6C_4 \times 2$$

$$k = \frac{{}^6C_4 \times 2}{{}^6C_5} = \frac{30}{6} \therefore k = 5.$$

7. A group consisting of two adults, two boys and two girls is to be seated at a round table. The adults are to be seated together. The boys and girls are to sit in alternating positions? How many different seating arrangements are possible?

2 Adults  
2 girls  
2 boys  
6

(A) 8

(B) 16

(C) 24

(D) 30

• Seat the first adult anywhere.

• 2nd adult has 2 ways.

• seat a first boy has 4 ways.

• 2nd boy has only 1 way.

• two girls sit 2 ways

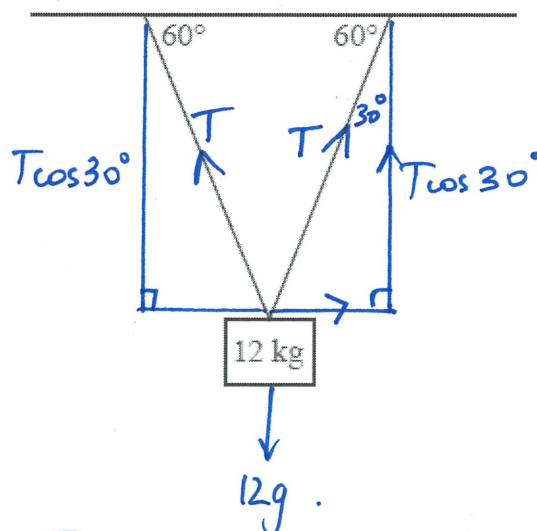
$\therefore$  seating arrangements

$$= 2 \times 4 \times 2$$

$$= 16.$$

8. A 12kg mass is suspended in equilibrium from a horizontal ceiling by two identical light strings. Each string makes an angle of  $60^\circ$  with the ceiling, as shown. If the force of gravity is  $g \text{ m/s}^2$  then the magnitude, in newtons, of the tension in each string is equal to:

- (A)  $12g$   
 (B)  $24g$   
 (C)  $4\sqrt{3}g$   
 (D)  $8\sqrt{3}g$



$$2T \cos 30^\circ = 12g$$

$$T = \frac{12g}{2 \cos 30^\circ}$$

$$T = \frac{12g}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 4\sqrt{3}g$$

9. The volume  $V$  of a spherical balloon of radius  $r$  mm is increasing at a constant rate of  $800 \text{ mm}^3$  per second. What is the rate of change of the radius with respect to time?

- (A)  $\frac{\pi r}{100} \text{ mm/second}$   
 (B)  $\frac{100}{\pi r} \text{ mm/second}$   
 (C)  $\frac{\pi r^2}{200} \text{ mm/second}$   
 (D)  $\frac{200}{\pi r^2} \text{ mm/second}$

$$\frac{dV}{dt} = 800 \text{ mm}^3/\text{second} \text{ . Find } \frac{dr}{dt} \text{ .}$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\therefore \frac{dr}{dt} = \frac{dV}{dt} \times \frac{dr}{dV}$$

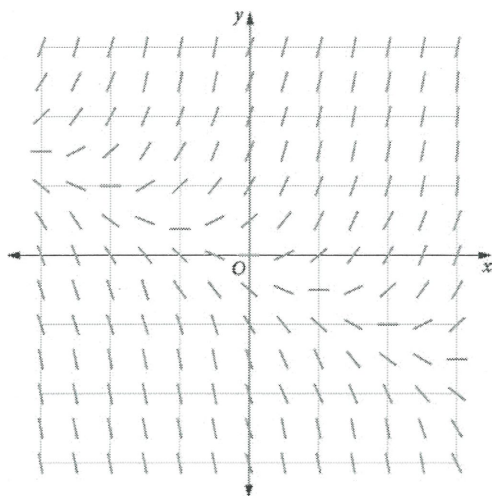
$$= 800 \times \frac{1}{4\pi r^2}$$

$$= \frac{200}{\pi r^2} \text{ mm/s}$$

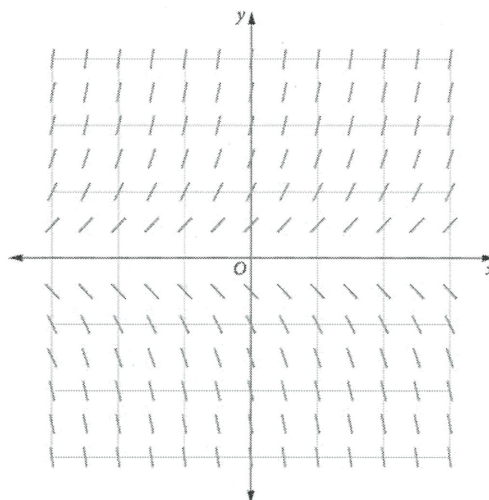
10. Which of the following direction fields could have the differential equation

$$\frac{dy}{dx} = x - ky \text{ as a solution if } k > 0 ?$$

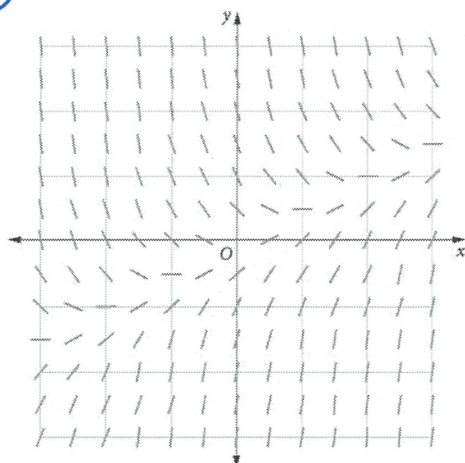
(A)



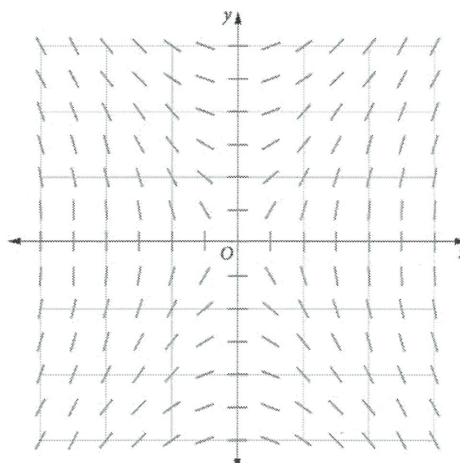
(B)



(C)



(D)



when  $\frac{dy}{dx} = 0$ :  $ky = x \Rightarrow y = \frac{x}{k}$  (line through origin, having horizontal line elements.  
 $\therefore$  not (B) and not (D).

But  $k > 0$ ,  $\therefore y = \frac{x}{k} \swarrow \therefore$  (C).

(a)

(i) Show that  $(x-3)$  is a factor of the polynomial

1

$$P(x) = x^3 - 4x^2 + x + 6.$$

(ii) Hence, express  $P(x)$  in factored form.

2

$$i) \quad P(3) = 3^3 - 4(3)^2 + 3 + 6$$

$$\therefore P(3) = 0. \quad \checkmark$$

①

$$ii) \quad \begin{array}{r} x^2 - x - 2 \\ x-3 \overline{) x^3 - 4x^2 + x + 6} \\ \underline{x^3 - 3x^2} \phantom{+ x + 6} \\ -x^2 + x \phantom{+ 6} \\ \underline{-x^2 + 3x} \phantom{+ 6} \\ -2x + 6 \\ \underline{-2x + 6} \\ 0 \end{array}$$

① for working if answer is incorrect.

$$\therefore P(x) = (x-3)(x^2 - x - 2)$$

$$\underline{P(x) = (x-3)(x-2)(x+1)} \quad \checkmark \checkmark \quad ②$$

(b) Evaluate  $\lim_{x \rightarrow 0} \left( \frac{\sin \frac{x}{3}}{4x} \right)$ 

1

$$\lim_{x \rightarrow 0} \left( \frac{\sin \frac{x}{3}}{4x} \right)$$

$$= \lim_{x \rightarrow 0} \left( \frac{\sin \frac{x}{3}}{\frac{x}{3}} \right) \times \frac{(\frac{x}{3})}{4x}$$

$$= 1 \times \frac{\cancel{x}}{3} \times \frac{1}{4\cancel{x}}$$

$$= \frac{1}{12} \quad \checkmark$$

①

(c) Solve:  $\frac{x^2-9}{3x} > 0$

2

$$\frac{x^2-9}{3x} > 0$$

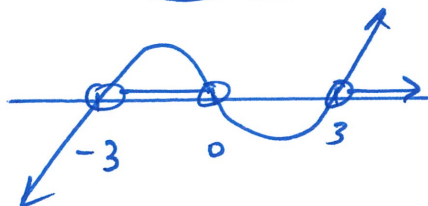
$$3x^2 \times \frac{x^2-9}{3x} > 0 \times 3x^2$$

$$x(x-3)(x+3) > 0$$

By observation:

$$\underline{\underline{-3 < x < 0 \text{ and } x > 3}} \quad \checkmark \checkmark$$

① for working  
if answer is  
incorrect



②.

(d) Evaluate:

2

$$\int \frac{2x}{\sqrt{25-x^2}} dx$$

$$\int \frac{2x}{\sqrt{25-x^2}} dx$$

$$= \int 2x \cdot (25-x^2)^{-\frac{1}{2}} dx \quad \checkmark$$

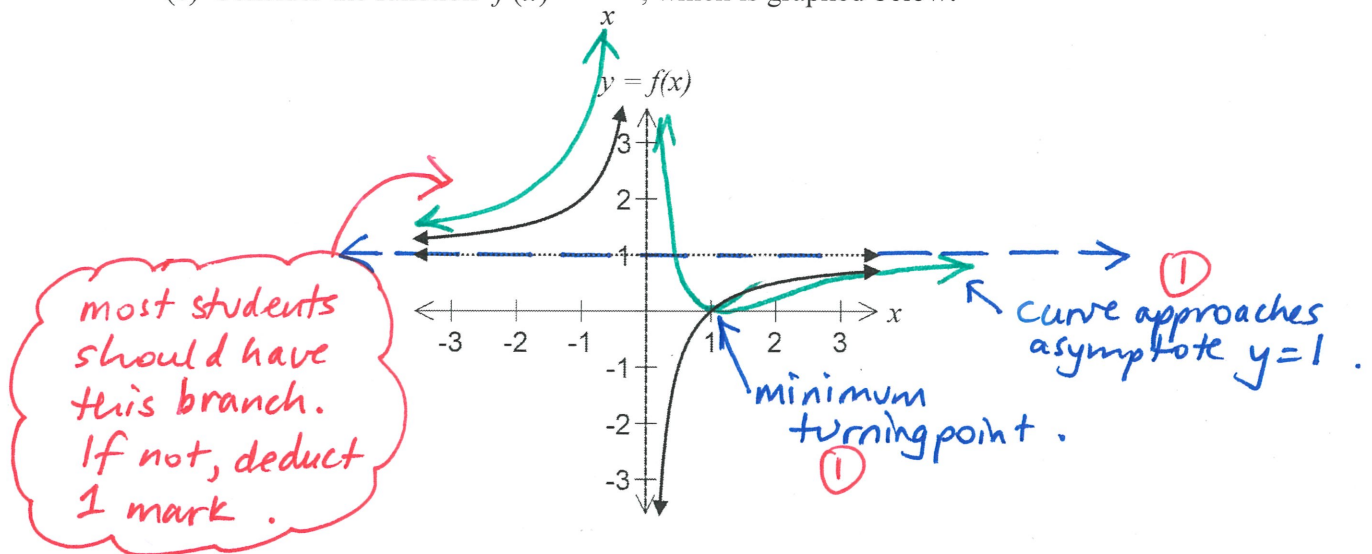
$$= -\frac{1}{\frac{1}{2}} (25-x^2)^{\frac{1}{2}} + C \quad (\text{reverse chain rule})$$

$$\underline{\underline{= -2\sqrt{25-x^2} + C}} \quad \checkmark$$

②.

**Question 11** (continued)

(e) Consider the function  $f(x) = \frac{x-1}{x}$ , which is graphed below:



Using the graph of  $y = f(x)$ , sketch the function  $y = \{f(x)\}^2$ .

2

**End of Question 11**

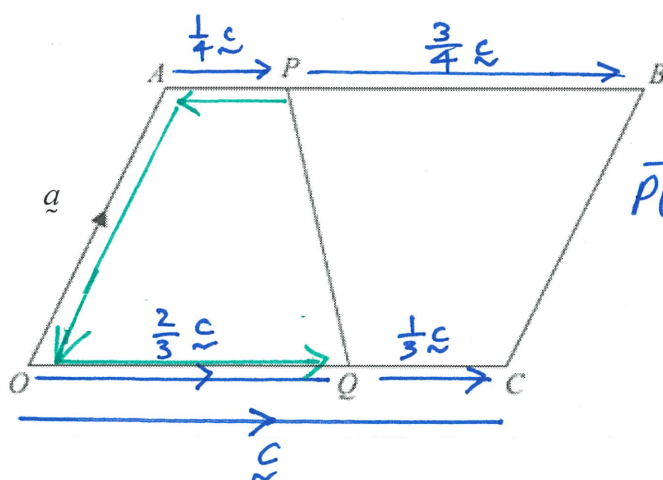
(a) In the diagram below,  $OABC$  is a parallelogram.  $\overrightarrow{OA} = \underline{a}$  and  $\overrightarrow{OC} = \underline{c}$

3

$P$  is the point on  $AB$  such that  $AP = \frac{1}{4}AB$ .

$Q$  is the point on  $OC$  such that  $OQ = \frac{2}{3}OC$ .

Find  $\overrightarrow{PQ}$  in terms of  $\underline{a}$  and  $\underline{c}$ , giving your answer in simplest form.



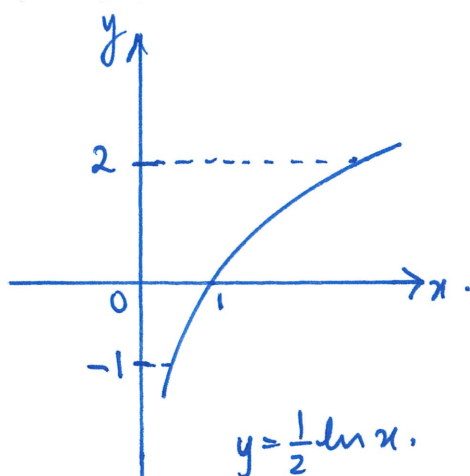
$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{PA} + \overrightarrow{AO} + \overrightarrow{OQ} \quad \checkmark \\ &= -\frac{1}{4}\underline{c} - \underline{a} + \frac{2}{3}\underline{c} \quad \checkmark \\ &= \left(\frac{2}{3} - \frac{1}{4}\right)\underline{c} - \underline{a} \\ &= \frac{5}{12}\underline{c} - \underline{a} \quad \checkmark \quad (3)\end{aligned}$$

(b) Find the volume, in terms of  $\pi$ , of the solid formed when the area

3

bounded by the curve  $y = \frac{1}{2}\log_e x$ , the lines  $y = -1$ ,  $y = 2$  and the  $y$ -axis

is rotated about the  $y$ -axis.



$$\begin{aligned}y &= \frac{1}{2}\ln x \\ 2y &= \ln x \\ e^{2y} &= x \quad \checkmark\end{aligned}$$

$$\begin{aligned}V &= \pi \int_a^b x^2 dy \\ &= \pi \int_{-1}^2 (e^{2y})^2 dy \quad \checkmark \\ &= \pi \int_{-1}^2 e^{4y} dy \\ &= \left[ \frac{e^{4y}}{4} \right]_{-1}^2 \times \pi \\ V &= \left( \frac{e^8}{4} - \frac{e^{-4}}{4} \right) \pi \text{ units}^3 \quad \checkmark\end{aligned}$$

(3)

**Question 12** (continued)

(c) Consider the expansion of  $(1+x)^{n-1}$  for integers  $n > 2$ .

(i) Show that:

$$n \binom{n-1}{1} + n \binom{n-1}{2} + \dots + n \binom{n-1}{n-2} = n(2^{n-1} - 2)$$

3

(ii) Find the smallest positive integer  $n$  such that:

$$n \binom{n-1}{1} + n \binom{n-1}{2} + \dots + n \binom{n-1}{n-2} > 5000$$

1

$$i) (1+x)^{n-1} = 1 + \binom{n-1}{1}x + \binom{n-1}{2}x^2 + \dots + \binom{n-1}{n-2}x^{n-2} + x^{n-1}$$

Let  $x=1$ :

$$(1+1)^{n-1} = 1 + \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} + 1 \quad \checkmark$$

$$2^{n-1} = 2 + \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2}$$

$$2^{n-1} - 2 = \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} \quad \checkmark$$

$$n(2^{n-1} - 2) = n \left[ \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} \right] \quad \checkmark$$

$$\therefore n \binom{n-1}{1} + n \binom{n-1}{2} + n \binom{n-1}{3} + \dots + n \binom{n-1}{n-2} = n(2^{n-1} - 2).$$

③

$$ii) \text{ Solve } n(2^{n-1} - 2) > 5000$$

Use trial and error:

$$n=8: 8(2^7 - 2) = 1008$$

$$n=9: 9(2^8 - 2) = 2286$$

$$n=10: 10(2^9 - 2) = 5100 \quad \therefore \underline{n=10} \quad \checkmark$$

①

**Question 13** Begin a new page.**(10 marks)**

- (a) A particle is moving in a straight line and is oscillating backwards and forwards. At time  $t$  seconds, it has displacement  $x$  metres from a fixed point  $O$  on the line, where  $x = A \cos\left(\frac{\pi}{4}t + \alpha\right)$ ,  $A > 0$ , and  $0 < \alpha < \frac{\pi}{2}$ . After 1 second the particle is 2 metres to the right of  $O$ , and after 3 seconds it is 4 metres to the left of  $O$ .

Given that  $A \cos \alpha - A \sin \alpha = 2\sqrt{2}$  and  $A \cos \alpha + A \sin \alpha = 4\sqrt{2}$ :

- (i) Solve the equations simultaneously, to find the values of  $A$  and  $\alpha$  in exact form. 2

- (ii) Show that the particle first passes through  $O$  after  $\frac{4}{\pi} \tan^{-1} 3$  seconds. 2

i)  $A \cos \alpha - A \sin \alpha = 2\sqrt{2} \dots \textcircled{1}$

$$A \cos \alpha + A \sin \alpha = 4\sqrt{2} \dots \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} : 2A \cos \alpha = 6\sqrt{2} \Rightarrow A \cos \alpha = 3\sqrt{2}$$

$$\textcircled{2} - \textcircled{1} : 2A \sin \alpha = 2\sqrt{2} \Rightarrow A \sin \alpha = \sqrt{2}$$

$$\therefore A^2 (\cos^2 \alpha + \sin^2 \alpha) = (3\sqrt{2})^2 + (\sqrt{2})^2$$

$$A^2 = 20$$

$$\underline{A = 2\sqrt{5}, A > 0.} \quad \checkmark$$

$$\frac{A \sin \alpha}{A \cos \alpha} = \frac{\sqrt{2}}{3\sqrt{2}}$$

$$\therefore \tan \alpha = \frac{1}{3}$$

$$\underline{\alpha = \tan^{-1} \frac{1}{3}} \quad (0 < \alpha < \frac{\pi}{2}) \quad \textcircled{2} \quad \checkmark$$

ii) when  $x=0$  :  $\cos\left(\frac{\pi}{4}t + \alpha\right) = 0$

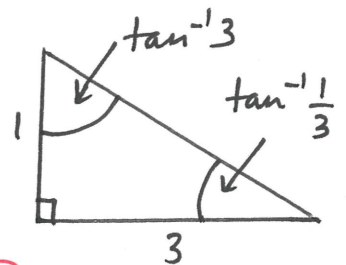
$t > 0$  :  $\frac{\pi}{4}t + \alpha = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

First such  $t > 0$  :  $\frac{\pi}{4}t = \frac{\pi}{2} - \alpha$  ①

$\frac{\pi}{4}t = \frac{\pi}{2} - \tan^{-1}\frac{1}{3}$

$\frac{\pi}{4}t = \tan^{-1}3$  ①

$\therefore t = \frac{4}{\pi} \tan^{-1}3$



**Question 13** (continued)

- (b) The position coordinates of any point on the path of a projectile at time  $t \geq 0$  in seconds, with initial velocity  $v \text{ ms}^{-1}$  at an angle of projection  $\theta$ , and acceleration downwards due to gravity,  $g$ , are:

$$x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - \frac{1}{2}gt^2$$

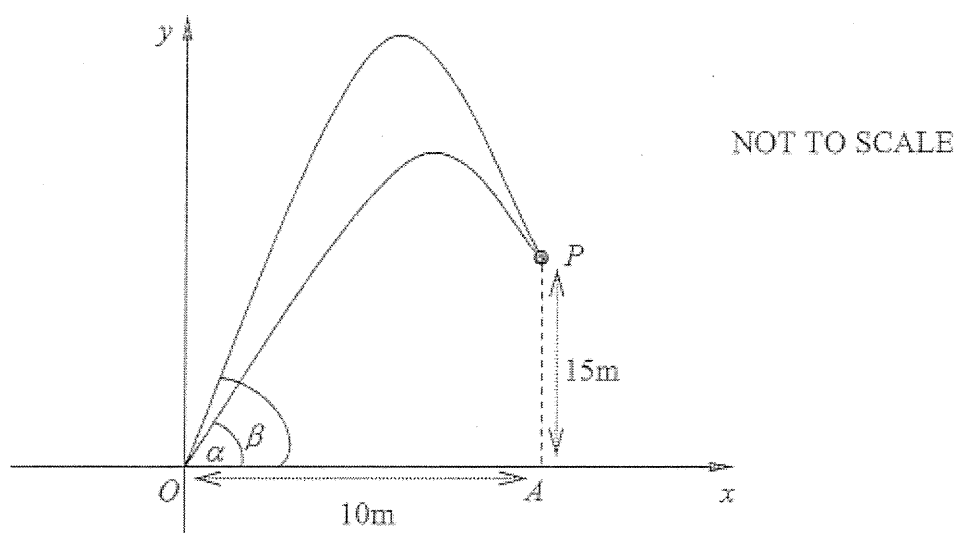
- (i) Show that the equation of the path of a projectile is given by:

$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

2

Harry throws a tennis ball from a fixed point  $O$  on level ground, with a velocity  $v = 7\sqrt{10} \text{ ms}^{-1}$  at an angle  $\beta$  with the horizontal. Shortly afterwards he throws another tennis ball from the same point at the same speed but at a different angle to the horizontal,  $\alpha$ , where  $\alpha < \beta$  as shown.

The two tennis balls collide at a point  $P$ , vertically above the point  $A$  on the ground, where  $OA = 10 \text{ m}$  and  $AP = 15 \text{ m}$ . The acceleration downwards due to gravity is  $g = 9.8 \text{ ms}^{-2}$ .



- (ii) Show that  $\tan \alpha = 2$  and  $\tan \beta = 8$ .
- (iii) Find, in surd form, the time elapsed between when the tennis balls were thrown.

2

2

**End of paper**

$$b) i) \quad x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - \frac{1}{2} g t^2$$

$$\Rightarrow t = \frac{x}{v \cos \theta} \quad \text{sub into } y:$$

$$y = (v \sin \theta) \cdot \frac{x}{v \cos \theta} - \frac{1}{2} g \left( \frac{x}{v \cos \theta} \right)^2 \quad \checkmark$$

$$y = x \tan \theta - \frac{g x^2}{2 v^2 \cos^2 \theta}$$

$$y = x \tan \theta - \frac{g x^2}{2 v^2} \sec^2 \theta \quad \checkmark \quad (2)$$


---

$$ii) \quad \text{At } P: x=10, y=15 \quad \text{and} \quad g=9.8, v=7\sqrt{10}.$$

$$\text{Using } i): \quad 15 = 10 \tan \theta - \frac{9.8(10)^2}{2(7\sqrt{10})^2} \sec^2 \theta$$

$$15 = 10 \tan \theta - \frac{9.8 \times 100}{2 \times 49 \times 10} (1 + \tan^2 \theta) \quad \checkmark$$

$$15 = 10 \tan \theta - 1 - \tan^2 \theta$$

$$0 = \tan^2 \theta - 10 \tan \theta + 16$$

$$0 = (\tan \theta - 8)(\tan \theta - 2)$$

$$\therefore \tan \theta = 8 \quad \text{or} \quad \tan \theta = 2 \quad \checkmark$$

$$\text{since } \alpha < \beta: \tan \beta = 8 \quad \text{and} \quad \tan \alpha = 2 \quad (2)$$

some progress

iii) Consider the two paths and find the time travelled to reach P:

Tennis ball 1:  $v = 7\sqrt{10}$ ,  $\theta = \beta$ ,  $\tan\beta = 8$  and  $x = 10$ :

$$t_1 = \frac{10}{7\sqrt{10} \cos\beta}$$

$$= \frac{10 \sec\beta}{7\sqrt{10}} \quad \left( \begin{array}{l} \sec^2\beta = 1 + \tan^2\beta \\ \sec^2\beta = 1 + 8^2 \end{array} \right)$$

$$= \frac{10 \times \sqrt{65}}{7\sqrt{10}}$$

$$t_1 = \frac{\sqrt{650}}{7}$$

✓ ① progress towards a solution.

Tennis ball 2:  $v = 7\sqrt{10}$ ,  $\theta = \alpha$ ,  $\tan\alpha = 2$  and  $x = 10$ :

$$t_2 = \frac{10}{7\sqrt{10} \cos\alpha}$$

$$= \frac{10 \sec\alpha}{7\sqrt{10}}$$

$$\left( \begin{array}{l} \sec^2\alpha = 1 + \tan^2\alpha \\ \sec^2\alpha = 1 + 4 \end{array} \right)$$

$$= \frac{10\sqrt{5}}{7\sqrt{10}}$$

$$= \frac{\sqrt{50}}{7}$$

$$\therefore \text{time elapsed: } t_1 - t_2 = \frac{\sqrt{650} - \sqrt{50}}{7} \text{ seconds.}$$

✓ ②.